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Paleomagnetic Study of La Quinta Formation, Venezuela[1]

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Abstract Eighty-seven oriented hand specimens of redbeds from the La Quinta Formation (Triassic-Jurassic), collected at four localities in the Venezuelan Andes, were studied. The results, after thermal demagnetization at 500°C, are moderately consistent at two of the four localities with normal and reverse polarization present. The average vectors are closer before correction for present bed attitude than after, a fact which suggests that the natural remanence of these rocks is of secondary origin. In some individual samples both normal and reversed components are present, which indicates that their magnetism was acquired during an interval that spanned at least one reversal of the earth's field. Comparison of the vectors erased on demagnetization and the residual vectors at the four localities suggests that the magnetism was acquired during the folding, and that its orientation is not related simply to either original or present bed attitude. Measured inclinations conform approximately with that of the present dipole field, but declinations are significantly northeast of it. If the declinations are valid, either a clockwise rotation of South America (for which there is no independent paleomagnetic evidence) or a tectonic rotation within the Venezuelan Andes would be implied.

Two samples of dacitic tuff (believed on tenuous grounds to be of Early Triassic age) were collected near the base of the La Quinta section at the type locality and yield results which are consistent with each other but which are distinct from those of the redbeds; the results from the dacitic tuff probably reflect the only primary magnetization measured in this study. After 150-oersted alternating field demagnetization and correction for bed attitude, their vector mean is $D = 13$, $I = -18$, $K = 106$, and $\alpha 95 = 25$. South coordinates are 68.5° S., 108.5° W.

Introduction

The La Quinta Formation is the basal unit of a thick sequence of unmetamorphosed sedimentary rocks of Mesozoic and early Cenozoic ages that lie unconformably on Paleozoic and older rocks in the Venezuelan Andes. The first paleomagnetic studies were made by Creer (1962, p. 157–159), who obtained widely scattered results from measurement of natural remanent magnetization (NRM) in orientated hand specimens collected at several localities. The most consistent orientations were obtained from outcrops believed to be near the type locality, where the beds that he sampled are almost flat-lying and generally gave poles with a southwest declination and negative inclination. Because no stability tests were performed, Creer refrained from assigning temporal significance to this orientation.

The Dirección de Geología of the Ministerio de Minas e Hidrocarburos, Venezuela, has instituted a comprehensive regional survey of the Venezuelan Andes and an overall report on the regional geology of the central part of the mountain belt is in preparation by Shagam. The magnetic measurements reported herein were made in the Rock Magnetism Laboratory at Princeton University, either by, or under the supervision of, Hargraves. Shagam is responsible for all field, mineralogic, and petrographic data.

The following abbreviations and other terminology are used in this report.

NRM	natural remanent magnetization
CRM	chemical remanent magnetization
TRM	thermoremanent magnetization
J	intensity of magnetization
n	number of specimens
D	declination of direction of magnetization
I	inclination of direction of magnetization
K	Fisher dispersion parameter (*see* Fisher, 1953)
$\alpha 95$	cone of confidence about direction of magnetization at 95 percent level of probability
AF	alternating field
oe	oersted

General Geology

The essential features of the La Quinta Formation and its tectonic setting in space and time are summarized.

Nomenclature and distribution.—The formation originally was named by Kündig (1938) to designate a redbed sequence of Mesozoic age

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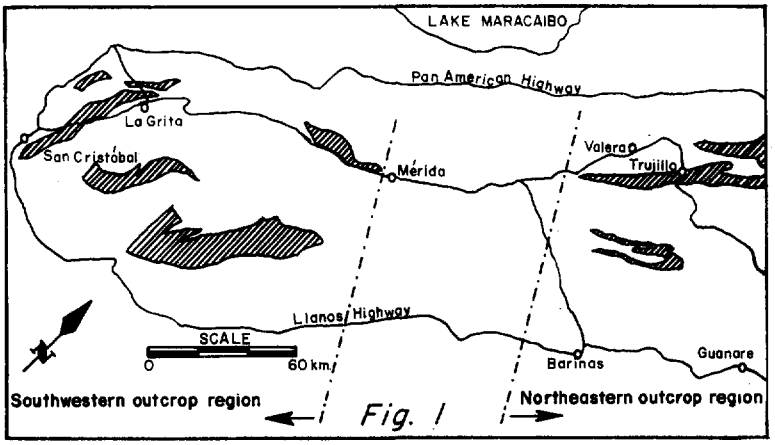


FIG. 1.—Generalized distribution of La Quinta Formation in Venezuelan Andes. Modified after Mapa Geológico de Venezuela, 1959.

(see discussion under "Age") that underlies marine Cretaceous rocks. Kündig defined a type section exposed on the La Grita-Seboruco road near the road junction of La Quinta. Rivero (1956) provided a summary of the work of early investigators.

Figure 1 depicts, in generalized form, the distribution of the formation in the Venezuelan Andes. As originally noted by González de Juana (1951), two distinct regions of outcrop, northeastern and southwestern, can be defined (Fig. 1). The map shows that rock sections assigned to the La Quinta Formation are in isolated areas of outcrop, but they are even more fragmentary than the map suggests. It has become common practice among Andean workers to assign to the formation all redbeds which can be shown or are believed to (1) overlie the fossiliferous marine sequence of the Palmarito Formation, the upper part of which is dated as probable Early Permian, and (2) underlie the Cretaceous section. The time interval between the Palmarito and Cretaceous formations could exceed 100 m.y., and it is clear that the isolated outcrop areas of the La Quinta Formation may comprise more than one redbed formation. The problem is further complicated by the known presence of a pre-Palmarito redbed formation of Carboniferous age (Sabaneta Formation). Where the Palmarito Formation does not crop out, it may be extremely difficult to distinguish Sabaneta rocks from those of the La Quinta.

The rocks for this study were collected from four stratigraphic sections (I, III, IV, V) in the southwestern outcrop region as shown in Figure 2. According to Rivero (1956) and as confirmed by the writers' observations, the type section (loc. III) is characterized by steeply dipping strata (60–70°). Creer (1962, p. 159), however, reported dips of about 10°. Nearly horizontal beds of the La Quinta Formation are known to crop out west of Seboruco, 6 km west of the type section. The location of Creer's specimens therefore does not correspond exactly to the writers' locality III.

Stratigraphy.—The La Quinta Formation is a thick sequence of massively bedded, dominantly coarse-grained rocks which are prominent cliff formers. In most sections, reddish colors are dominant but interbeds of pale to dark gray and green are not uncommon. The sections change significantly from area to area (*cf.* Figs. 3–6) and no subdivision into members of regional extent is possible.

Lithology and petrology of oriented specimens.—Most specimens are within the range of fine-grained red siltstone to medium-grained sandstone; a few coarse-grained sandstone and conglomerate specimens also were studied. Most of the specimens are purple-red and most are laminated or show other evidence of stratification. Cross-lamination of simple type is present in some specimens, but is not notably well developed. A third of the specimens show evidence of penecontemporaneous deformation in the form of small-scale slump structures, mudcracks, intraformational mud chips, and blotchy textures produced by burrowing organisms.

In thin section most specimens show a poorly sorted texture with clastic grains of

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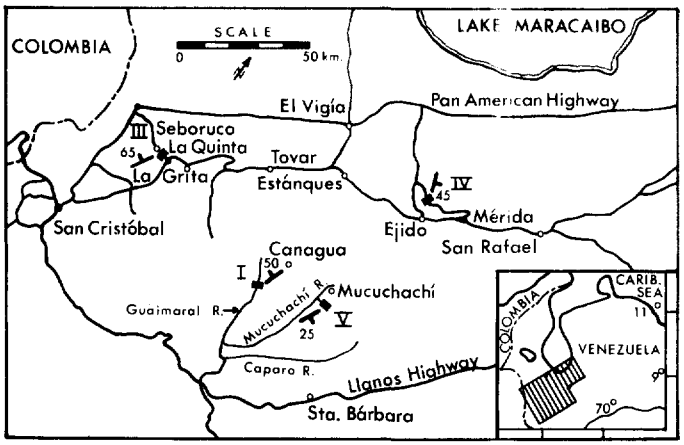


FIG. 2.—Geographic locations of measured stratigraphic sections I, III, IV, and V, La Quinta Formation, Venezuela. Structure symbols represent regional strike and dip at each locality.

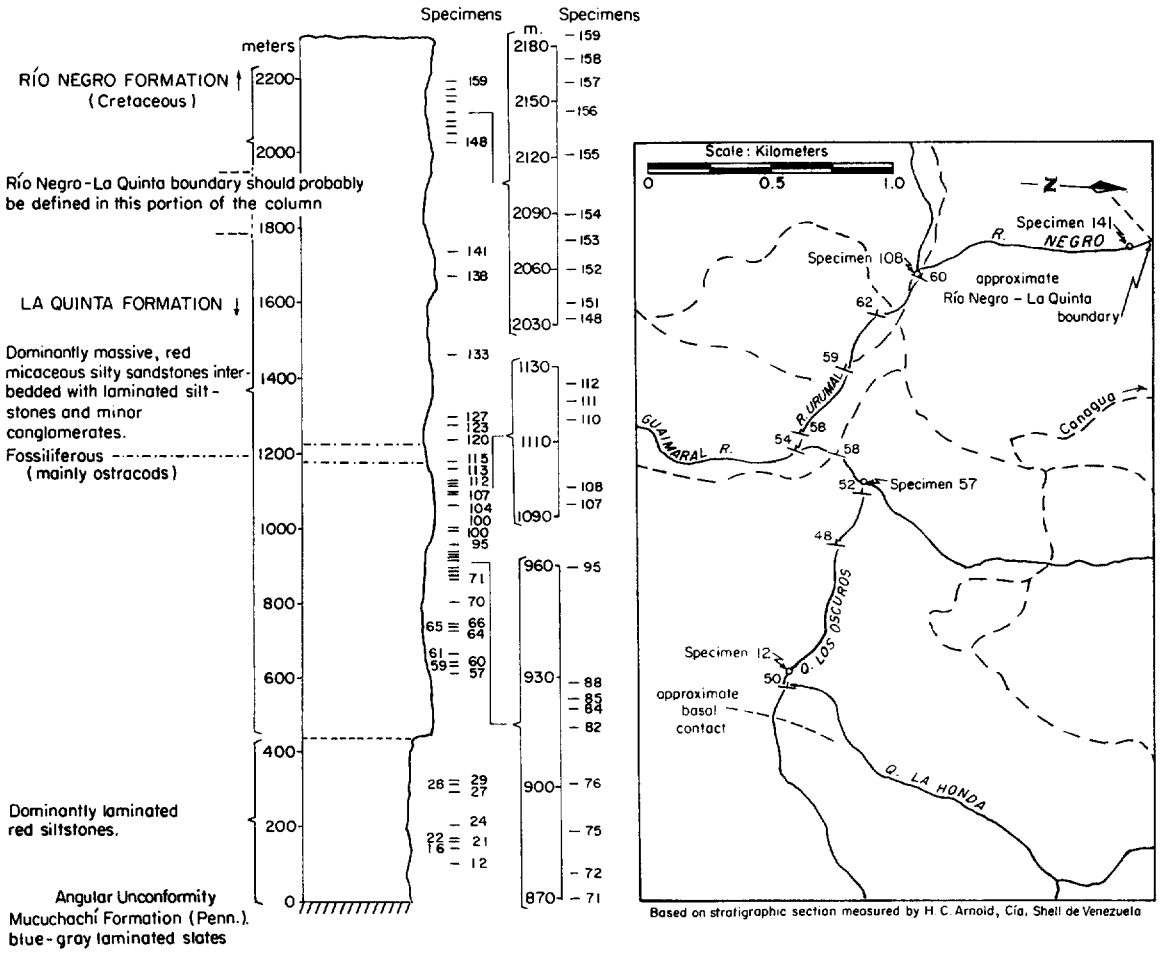


FIG. 3.—La Quinta Formation, Río Guaimaral area, Canagua, Mérida State, Venezuela. Sample area I.

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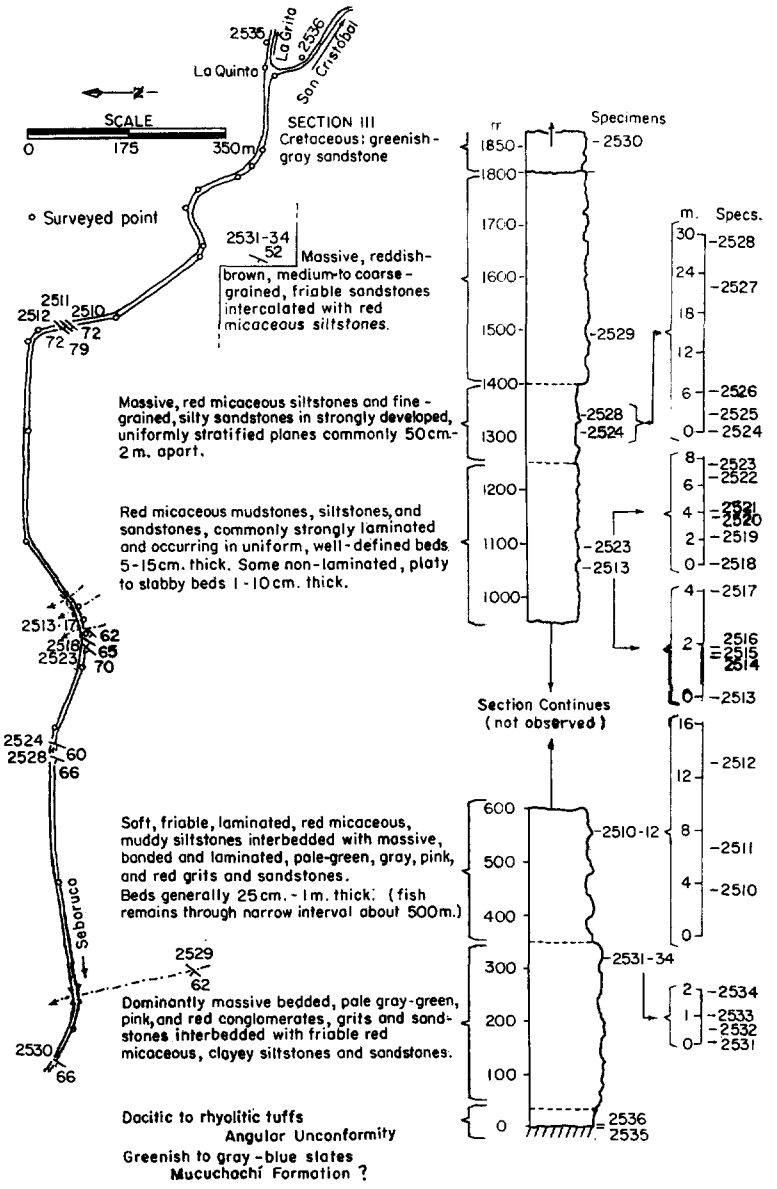


FIG. 4.—La Quinta Formation type section, La Quinta, Táchira State, Venezuela. Sample area III.

quartz and subsidiary microcline, plagioclase, mica, and chlorite embedded in a red-stained clayey matrix which is locally calcareous. Many specimens show a marked development of secondary silica overgrowths on the quartz grains.

Opaque oxides.—Opaque oxides are present as detrital grains, as a coating on other detrital grains, as matrix element or void fillings, and as alteration products.

The clastic opaque grains invariably prove, on optical examination, to be hematite. This was confirmed by X-ray powder photographs of the opaque grains from five specimens. Coatings of iron oxide on clastic grains are believed to be hematitic on the basis of their red color. The coatings were emplaced before suturing and cementation, as indicated by their presence where adjacent grains are closely molded, and at contacts of secondary silica with host grains.

The dark reddish-brown, semi-opaque mate-

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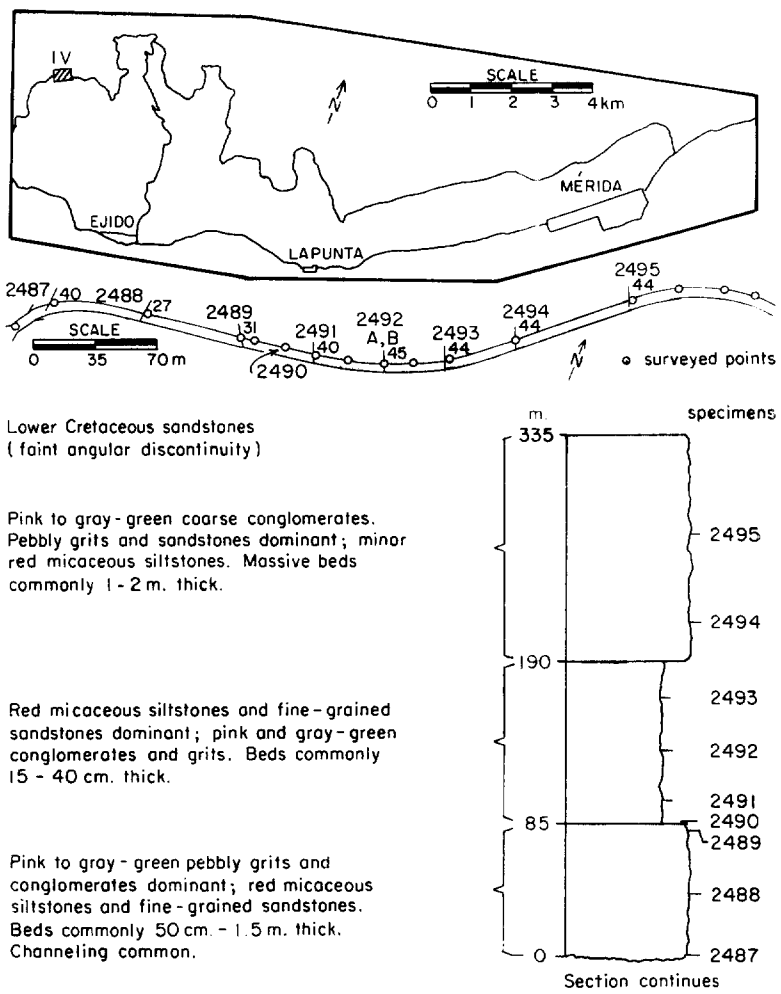


FIG. 5.—La Quinta Formation, partial section exposed along Pan American Highway about 5 km northwest of Egido, Mérida State. Sample area IV.

rial of the matrix and void fillings reaches a maximum of about 15 percent by volume in the rocks of darkest color and is almost absent from gray-green specimens. Thus, although overall rock color appears to be related primarily to the amount of reddish-brown material present, the latter is not the dominant constituent of any of the oriented specimens. Initial attempts to make positive identification of this material have not been satisfactory. Iron oxide also takes the form of very thin, pale-yellow, limonitic stains on all other grains. This alteration is clearly of secondary origin, of small volume, and generally restricted to fractures and joints. It is not believed to contribute to the paleomagnetic properties of the rocks.

Environment of deposition.—On the basis of regional mapping, it is inferred that uplift of the central part of the Venezuelan Andes led to the development of elongate structural troughs extending northeast and southwest from the central core. Great thicknesses of relatively coarse, nonmarine detritus were poured into the troughs. The tectonic setting may have been like that pictured by Krynine (1950) for the Triassic sedimentary rocks of Connecticut (Newark Group).

Age.—Rivero (1956) has reviewed the evidence concerning the age of the La Quinta Formation. Fish remains, apparently of the genus *Lepidotus,* have been found in the lower part of the section at the type locality and were

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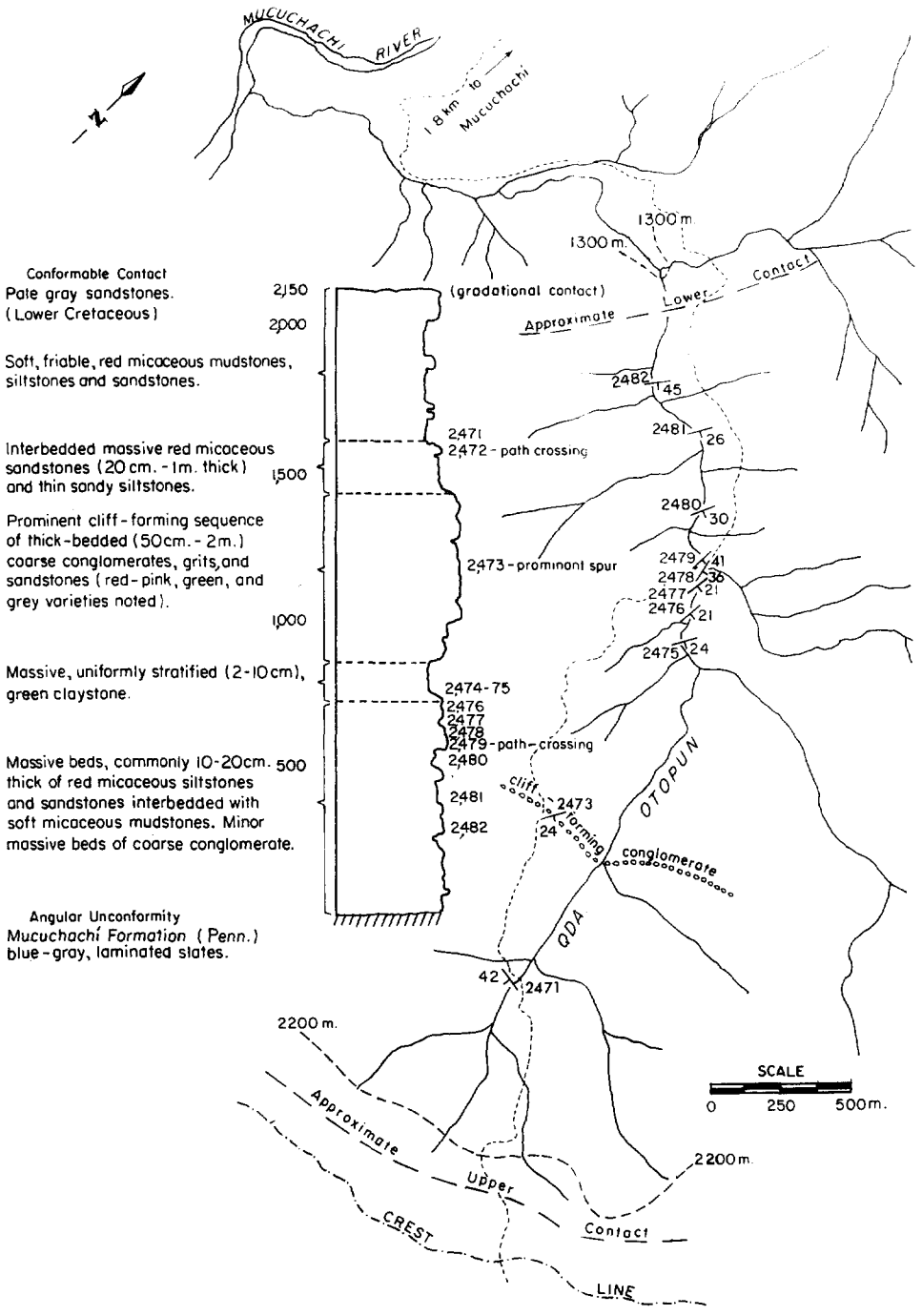


FIG. 6.—La Quinta Formation, Río Otopún section, Mucuchachí, Mérida State. Sample area V.

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considered by Woodward (*in* Kündig, 1938, p. 33) to be of probable Late Triassic to Early Jurassic age. Bermúdez (1962) identified two species of crustaceans in a specimen collected by L. A. González from the Guaimaral locality (I) as *Howellites colombianus* Bock and *Isaura olsoni* Bock. These and a similar fauna containing other species of *Howellites* and *Isaura,* collected from the Lockatong Formation (in the Newark Group) of New Jersey, are considered to be of Late Triassic age by Bock (1953).

One tenuous line of evidence concerning the age may be of significance—a youngest Rb/Sr age of approximately 230 m.y. (*i.e.,* roughly equivalent to the Permian-Triassic boundary on the Kulp scale, 1959) obtained from biotite in a schist specimen of the Andean basement (Bass and Shagam, 1960). These rocks are known to be of pre-Pennsylvanian age and probably of pre-Middle Ordovician age on the basis of regional mapping. However, no episode of regional metamorphism of Permo-Triassic age can be defined inasmuch as the Permo-Pennsylvanian formations (Sabaneta and Palmarito) have not been metamorphosed regionally. It is conceivable that the 230-m.y. age reflects a thermal event related to a major episode of granite intrusion coeval with the siliceous tuff at the base of the section at the type locality (III).

The definition of the minimum age limit of the La Quinta Formation is complicated by later geologic events (Fig. 7). Deposition of the La Quinta sediments was followed by a marine transgression from north to south, during which the deep Uribante trough was developed over the southwestern part of the modern Andes (*cf.* van Andel, 1958). The basal unit of the marine facies, the Río Negro Formation, is composed largely of nonfossiliferous sandstone. Ages obtained from minor interbeds of fossiliferous limestone and shale suggest an Early Cretaceous age (Neocomian-Barremian) for most of the formation (Trump and Salvador, 1964), but the age may extend into the Jurassic. The fact that the uppermost specimens at locality I have been assigned to the Río Negro Formation suggests that they are of Early Cretaceous age, but they could be older. Because isochronous lines cannot be defined on Figure 7 below the Apón limestone, chronostratigraphic correlation between La Quinta sections is not now possible.

Within the limitations described, the following essential evidence bears on the age of the La Quinta Formation.

1. The fact that three of the four sections grade transitionally upward into rocks of Early Cretaceous age suggests that at least the upper parts of these sections are of Late Jurassic to Early Cretaceous age.

2. Should the faunal determination prove to be correct, then to a first approximation the parts of the stratigraphic columns of localities I and III above the faunal zones may be considered to be of Jurassic age.

3. If the argument based on radiometric data is assumed to be correct, the age of the tuff at the base of the section at the type locality (Fig. 4) possibly is latest Permian or earliest Triassic, and the redbeds between the tuff and the fossiliferous beds would constitute a Triassic part of the stratigraphic column. No structural discontinuity was noted between the tuff and the sedimentary rocks, but the contact might represent a considerable time gap.

4. At locality IV (near Egido), the section does not grade conformably into the Cretaceous sequence and the contact is slightly angular. Moreover, neither fossiliferous beds nor volcanic tuffs have been identified. Hence, little can be said about the age range of the section.

Structural geology.—Most of the structural deformation of the post-Paleozoic section was in the form of uplift which resulted in the formation of monoclinal tilt blocks. In contrast, pre-Mesozoic rocks show evidence (tight fold-

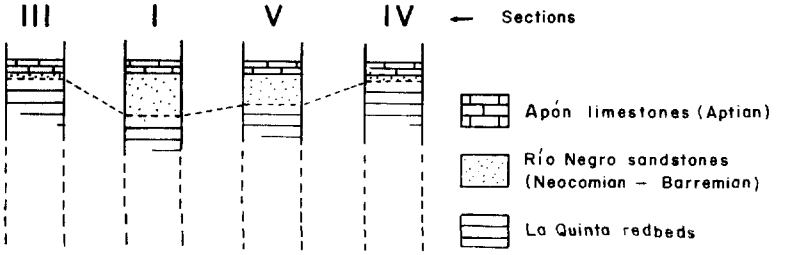


FIG. 7.—Generalized relations between La Quinta Formation and overlying marine Cretaceous rocks.

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ing, well-developed cleavage, *etc.*) indicative of lateral tectonic transport.

The four stratigraphic sections sampled are in four distinct tilt blocks. The structural orientation at each locality can be read from the maps of Figures 3–6, and the overall structural relations among the four localities are shown in Figure 2. Spatial relations indicate that the tilting must have involved a considerable component of vertical movement. On the basis of stratigraphic thicknesses and present elevations of the formations, and if late or post-Eocene uplift is assumed, it can be shown that the vertical uplift was about 5–6 km for the Guaimaral locality and probably 1–2 km less for localities III and V.

A study of slickensides on stratification planes of specimens from the Guaimaral locality (I) suggests that interbed slip may have been a significant part of structural movement. Whereas slip in the direction of true dip is commonly indicated, almost 40 percent of the specimens show evidence of slip at an angle of 25–35° from the direction of true dip. This may reflect a measure of triclinic symmetry in the tilting movements or, alternatively, subsidiary rotations about axes normal to bedding. Regardless, these factors may provide part of the explanation for the dispersion of paleomagnetic vectors plotted in Figure 9.

In the vicinity of locality IV (near Egido) the post-Paleozoic section forms a complex series of small fault-bound blocks believed to have formed by gravity tectonics. No significant change in declination from that at localities I and III is evident from the paleomagnetic data (Fig. 9), but there does appear to be a significant difference in inclination.

Inasmuch as the Cretaceous rocks conformably overlie three of the measured sections, it is clear that tilting took place after deposition of the Cretaceous rocks. Approximately 30 km south of the Guaimaral and Otopún localities, along the Río Caparo, rocks of Paleocene to Eocene age appear to be conformable on the Upper Cretaceous section. Similar relations are known in the vicinity of Mérida, San Cristóbal (Trump and Salvador, 1964), and elsewhere in the Venezuelan Andes. After Eocene time, most of the mountain belt formed a topographic high, and younger strata are present only as an apron around the northern and southern foothills of the mountains.

Clearly a major tectonic uplift occurred approximately at the end of the Eocene. Thick, coarse-grained deposits of Miocene and Pliocene ages form the dominant part of the sedimentary apron, but evidence is insufficient to show whether these deposits are related to a single end-of-Eocene uplift, to two or more distinct pulses of uplift in post-Eocene time, or to continuous uplift which extended well into middle or late Cenozoic time. Another distinct pulse of uplift did occur in late Pliocene to Pleistocene time as shown by (1) the presence of thick terrace gravel within the mountain belt (the deposition of the gravel was related to tectonic uplift), and (2) such geomorphologic evidence as deep incision of young erosion surfaces. Clearly a minimum of two "pulses" was involved in the tilting of post-Paleozoic rocks.

Tectonic setting.—The most striking feature of the uplift of the Venezuelan Andes is that it was a "cold" deformation (in the sense that no thermal event is recorded). On the northeast (Coast Ranges of Venezuela-Caribbean Island arc) and southwest (main Andean ranges), post-Paleozoic history, in marked contrast, is notable for significant thermal events—either combined igneous activity and regional metamorphism (Caribbean), or igneous activity alone (Andes). The writers therefore postulate that the basic mechanism of uplift and tilting in the Venezuelan Andes was related less to deep-seated events in the mantle beneath those ranges than to the product of mechanical reaction between crustal blocks. One possible mechanism is suggested by the much-cited concept (*e.g.*, Hess, 1962) that subcrustal convection causes crustal drift, and more specifically the concept that South America was drifting westward relative to the Caribbean during the Cenozoic. Such relative movements of large crustal blocks could have given rise to the stress field shown in Figure 8, part of which (*a-a'* and *b-b'*) originally was proposed by Bucher (1952, p. 26) to explain major structural elements of the Venezuelan Andes. One possible consequence of such a stress field, significant in the light of the paleomagnetic results reported herein, is clockwise rotation of the Andes if the stress were operative during a long period of time instead of during a very short period (*i.e.*, a "momentary" pulse).

Field and Laboratory Procedures

Sampling.—Oriented hand specimens were collected at intervals throughout the La Quinta section at localities I, III, and V, and from the upper part of the section at locality IV. The

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numbers of the samples and the corresponding stratigraphic sections are shown in Figures 3–6.

The specimens from locality I (Guaimaral region) were collected by L. A. González using as a stratigraphic reference a section measured by H. C. Arnold, Compañía Shell de Venezuela, at a scale of 1:1,000. Because the section shows a relatively monotonous sequence of dominantly massive sandstone, it was not possible to determine the absolute stratigraphic level of each specimen. However, by use of such reference points as stream tributaries, path crossings, waterfalls, *etc.*, marked on the section, the absolute stratigraphic locations indicated in Figure 3 are believed to be accurate to within ±40 m.

At localities III (La Quinta, type section) and IV (near Egido), the sections are exposed along highways and were measured by pace-and-compass methods under optimum conditions. The stratigraphic location of specimens (Figs. 4, 5) is believed to be correct to within ±40 m. At locality V (Río Otopún) pace-and-compass methods also were employed but under difficult conditions, there being almost 1,000 m of relief between the base and top of the section. In this section, both total thickness and the absolute stratigraphic location of specimens are imperfectly known (the latter at best to within ±100 m).

In summary, at localities I and V all specimens were collected at or near the surface, mainly from stream outcrops. At localities III and IV most specimens were collected from road cuts at depths ranging from a few meters to more than 10 m below the former natural land surface, but the outcrops had been exposed in the cuts for a minimum of 3 years. At both localities explosives had been used to make the cuts.



FIG. 8.—Possible crustal stress field in northern South America and Caribbean. a-a′ = compressive stresses produced by b-b′ = shear couple related in turn to c-c′ = relative crustal drift directions.

Laboratory methods.—Cores 0.98 in. in diameter were drilled from the hand specimens in a drill press and oriented in a sand box. Remanence measurements were made on a spinner magnetometer (Phillips and Kuckes, 1967). In alternating-field (AF) demagnetization procedure (McElhinny, 1963), the specimen is spun at 160 rpm using a 2-axis system with a 16:15 ratio. The alternating field is brought up to the peak value and down very slowly with an inductrol. In thermal demagnetization (Phillips, 1964), the specimen is heated in a furnace inside Helmholtz coils and cooled in field-free space ($<50\gamma$). An average of three separate specimens (0.98 in. diameter by 0.9 in. length) were measured from each hand specimen. As is common, alternating-field demagnetization was found to have little effect, and was studied only at a reconnaissance level. After exploratory thermal demagnetization at small increments of temperature, all specimens were thermally demagnetized at 500 and 600°C. Most also were measured after 400°C.

Samples which gave inconsistent results such that the Fisher dispersion parameter (K) was less than 4.0 were omitted from further consideration. Seventy-three of the 87 samples survived this test.

Paleomagnetic Results

Remanence.—The results obtained are illustrated in Figure 9. Despite considerable scatter, a clustering of directions with northeast and southwest declinations is evident at locality I, and a northeast cluster is seen at locality III. At both, however, miscellaneous directions also are obtained. The results from localities IV and V show a wider scatter than those from localities I and III.

For purposes of averaging and statistical analysis the directions were grouped arbitrarily as shown in Figure 9. These results are presented in Table I and the locality group means are shown in Figure 10. Samples III-35 and III-36 are volcanic tuff collected at the base of the La Quinta section at the type locality and are grouped separately in Table I and Figure 10.

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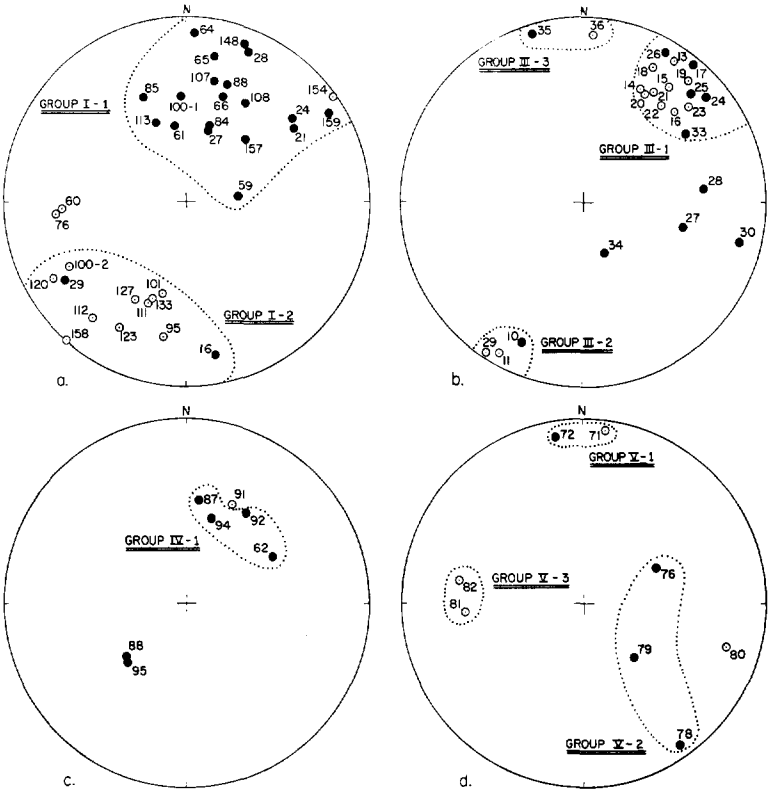


FIG. 9.—Lambert meridional equal-area projections of average remanent vectors of individual samples at localities I = a, III = b, IV = c, and V = d, after thermal demagnetization to 500°C, and before correcting for bedding attitudes (Table I). Dotted line in each indicates arbitrary grouping of vectors, means of which are shown in Figure 10 and Table I. Circles = negative inclination (above horizontal) of magnetic directions plotted on upper hemisphere; dots = positive inclination plotted on lower hemisphere.

Table I. Summary of Paleomagnetic Data Obtained from La Quinta Samples at Four Localities[1]

Locality	*Samples Studied*	*Samples Used*	*Group*	*Samples in Group*	*NRM*								*Samples*	*500°C*							
					Uncorrected				*Bed corrected*					*Uncorrected*				*Bed corrected*			
					D	*I*	*K*	α95	*D*	*I*	*K*	α95		*D*	*I*	*K*	α95	*D*	*I*	*K*	α95
I	41	35	I-1	25	34	41	11	9	348	38	9	10	20	23	39	8	13	348	26	7	15
			I-2	9	210	24	9	19	184	18	8	20	12	216	−18	6	20	190	−22	6	19
III	27	23	III-1	15	32	1	7	16	32	2	6	18	15	36	−11	11	12	44	2	8	14
			III-2	3	216	40	12	37	248	6	10	42	3	210	5	39	20	213	0	28	24
			III-3	2	4	4	93	26	14	15	93	26	2	1	5	107	24	13	18	106	25
IV	9	7	IV-1	4	72	64	4	55	76	22	3	60	4	64	47	34	7	67	7	77	11
V	10	8	V-1	2	357	26	—	—	342	60	—	—	2	3	11	17	64	358	45	—	—
			V-2	4	87	47	3	70	87	20	3	64	3	121	44	4	72	113	18	4	68
			V-3	2	296	−2	—	—	293	34	—	—	2	273	−33	78	29	278	− 1	20	59

[1] Values of *NRM* and measurements following demagnetization procedures (150 oe. A. F. for the volcanic tuff samples constituting Group III-3; thermally to 500°C for all others). Vectors at each locality have been grouped arbitrarily as indicated in Figure 9. See introduction for key to abbreviations.

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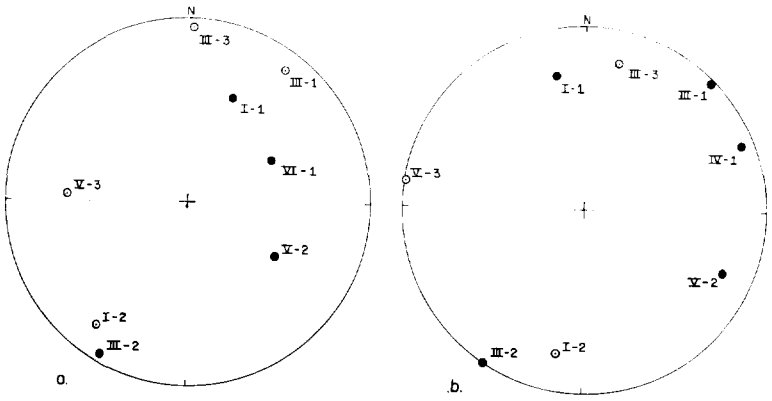


FIG. 10.—Lambert meridional equal-area projections of average vectors from each locality group, (a) before, and (b) after, correcting for bed attitudes. Note that agreement between Groups 1 and 2 from localities I and III, although poor at best, is better before correcting for bedding attitude. This evidence suggests that magnetization is secondary. Key to circles and dots as in Figure 9.

The average vectors at sites I and III (which give the best results) correlate better, though poorly at best (Fig. 10), before correction for bed attitude, which implies that the remanence measured is secondary and was acquired after folding.

Remanence intensities and stability.—The natural remanence intensities of the samples measured range from 5 to 0.1 $\times$ 10^{-5} emu/cc. During thermal demagnetization (Fig. 11) most specimens show little change up to 400°C, and a slight to moderate decline at 500 and 600°C. Some specimens show a very great increase in intensity above 600°C. Whether this is the result of the acquisition of thermoremanent magnetization (TRM) related to incomplete cancellation of the field, or to the selective demagnetization of one of two mutually opposed vector components, is not clear. Because most specimens in a thermal experiment behave normally, it is thought unlikely that TRM has been acquired. However, there is no consistency in the orientation of the remanent vector in these stronger samples. Some samples show a striking increase in bulk susceptibility after heating to 700°C, but this change does not correlate in every case with the increase in *J* (intensity).

The thermal-stability characteristics correlate roughly with the lithologic character of the specimen. Fine-grained red siltstone is more strongly magnetized and relatively stable. Coarser grained, more conglomeratic samples (as from loc. IV) are weaker at the beginning and decline steadily during demagnetization. The within-sample vector dispersion correlates with average intensity, stronger samples showing higher *K* values.

Vector erased on demagnetization.—Samples from locality I are in two distinct groups of op-

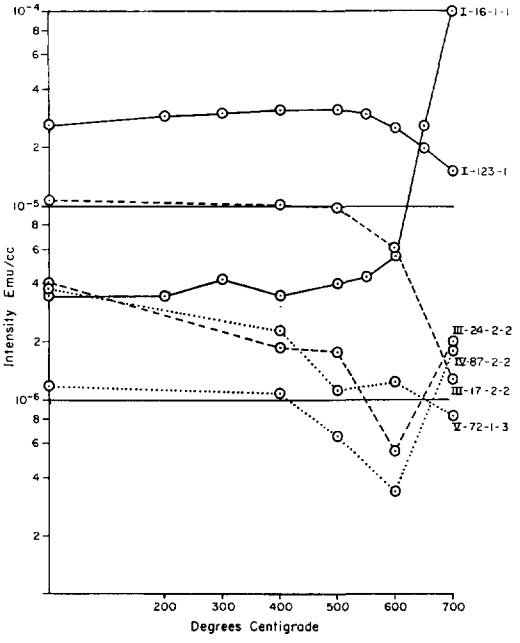


FIG. 11.—Variations of *J* (intensity of magnetization) with temperature on demagnetization. Specimens from different localities selected to illustrate diverse behavior; note spectacular increase of intensity (of unknown origin) during heating of some samples to 700° C and cooling in field-free (<50γ) space.

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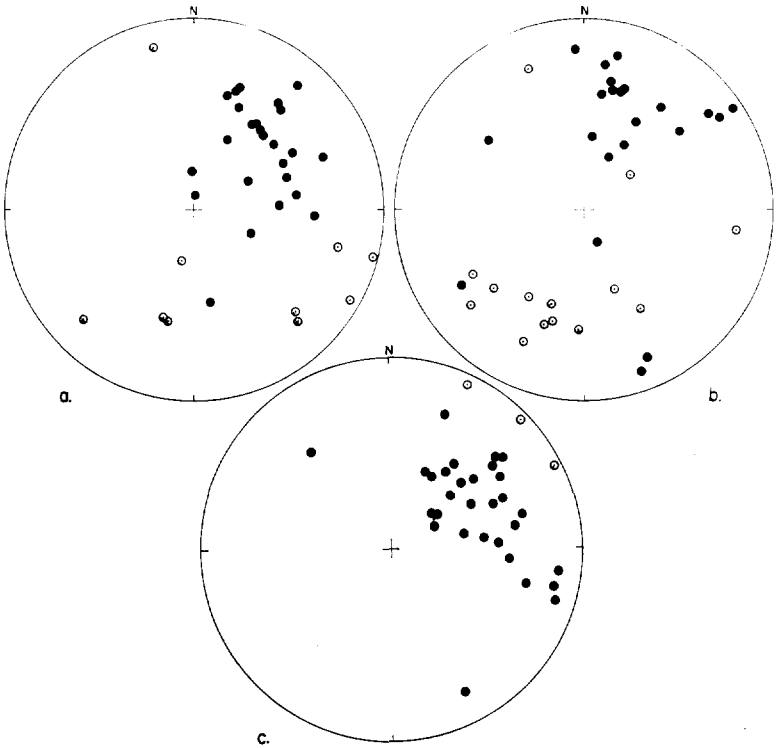


FIG. 12.—Equal-area projections comparing (a) NRM vector, (b) residual vector after 500°C thermal demagnetization, and (c) vector erased by heating to 500°C. ($\overline{\text{NRM}} - \overline{500}$°C.) Results from set of individual specimens from locality I are shown. Note relative consistency of erased vector. Key to circles and dots as in Figure 9.

posite orientation (Fig. 9). Whereas samples with northeast declination decreased in intensity during demagnetization, those with southwest declination increased. This implies that an unstable component with northeast declination was being erased from both groups. With the aid of a computer, the vector erased during demagnetization was calculated for all data and the results on a selected suite of specimens are shown in Figure 12.

The orientation of the vector obtained by subtracting the residual vector from the original is extremely sensitive to the magnitude of the vectors. Hence, to have any vectorial significance the relative precision of the intensity measurements must be high, especially if the two vectors being subtracted are of similar magnitude.

The data illustrated in Figure 12 were obtained under optimum circumstances (same operator and instrument, and measurements made on successive days). Only one specimen per sample is represented. The measurements on other specimens from these samples were made at different times, and at widely spaced intervals of time. The degree of internal consistency is not high. The vector erased from all specimens has been analyzed and averaged. The samples in which the dispersion parameter of the vector erased was less than four have been omitted. The resulting average vectors are plotted in Figure 12, and the data are listed in Table II.

Comparison with previous data.—The samples measured by Creer (1962) give NRM values which are predominantly reversed and have a southwest declination. The writers' readings for the type locality (III) have a predominantly northeast declination (Fig. 9) and only three are reversed. Mixed polarities were found in greater abundance at locality I, but on the whole inversions of polarity are not believed to have time-stratigraphic significance.

Interpretation of Paleomagnetic Data

Discussion.—The mean vector erased from

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Table II. Summary of Average Vector Erased by Thermal Demagnetization Between NRM and 500°C, and Between 500°C and 600°C, for Samples from Four Localities[1]

		NRM—500°C									500°C–600°C							
		Uncorrected				Bed corrected					Uncorrected				Bed corrected			
Locality	n	D	I	K	α95	D	I	K	α95	n	D	I	K	α95	D	I	K	α95
I (Specs. in Fig. 12)	35	63	35	11	11	0	53	6	15	24	275	−21	7	12	254	−71	7	12
III	19	19	32	4	21	355	5	4	20	20	12	− 7	3	25	31	−17	3	25
IV	8	20	72	4	31	53	41	5	29	8	40	−18	2	61	26	−55	2	60
V	8	357	35	4	31	16	40	3	39	10	70	−10	2	46	64	−23	2	45

[1]Summary data on individual specimens from samples at locality I (Fig. 12) are also included. See introduction for key to abbreviations. "*n*" is number of samples averaged.

NRM by heating in field-free space to 500°C is, at locality I, indistinguishable from the mean residual northeast vector (group I-1). However, this vector is erased from both groups I-1 and I-2. Thus it is clear that the NRM of the group I-2 samples is composite and includes both a normal component with northeast declination and a reversed component with southwest declination, the latter being the more stable.

At this locality groups I-1 and I-2 are almost diametrically opposed. Thus these data indicate only that the measurable natural remanence of the group I-2 samples was acquired during a time interval of sufficient duration to span a reversal of the earth's field. Whether this was soon or long after deposition remains to be established.

The vectors erased from samples from locality V are significantly better grouped than the NRM or the residual vectors from those samples. This suggests that a more stable component is present which has a different orientation than the vector erased, the two being of different relative magnitudes in different samples. Because the erased vectors group somewhat better before correction for bed attitude (Fig. 13), the writers conclude that the predominant to exclusive magnetic vector in all La Quinta redbeds is secondary, acquired during a time interval which "straddles" at least one field reversal, subsequent to the folding of the beds. The fact that the vector erased from samples from locality IV (*after* bedding correction) compares better with those from other localities *before* bedding correction (Fig. 13) suggests that the magnetism of the La Quinta may have been imposed during the periods of deformation, being "fixed" at some places before folding and at others after folding.

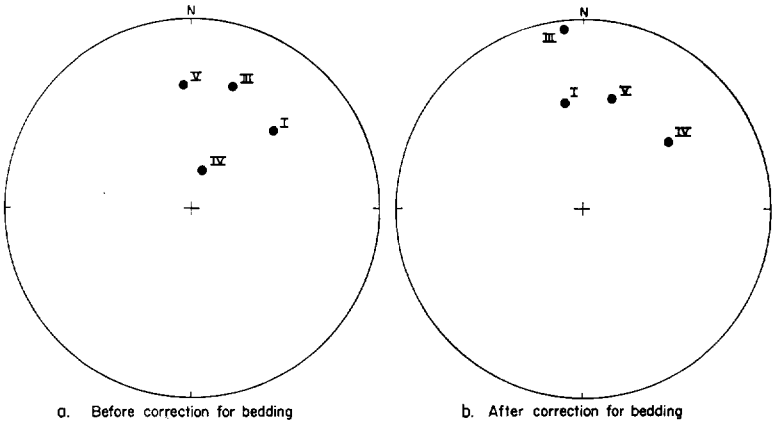


FIG. 13.—Lambert meridional equal-area projection of average vector erased by thermal demagnetization to 500°C of samples from all four locations. (a) Before correction for bedding, (b) after correction for bedding. Note that vectors for localities I and III, before correcting of bedding attitude, agree better with that for locality IV after bed correction. This implies that secondary magnetization was acquired more or less contemporaneously, before rotation in some areas, and after rotation in others.

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Table III. Summary Data from Stratigraphic Subgroups at Locality III from Base of Exposed Section Upward (1–6)[1]

Samples	NRM								500°C								Subgroup
	Uncorrected				Bed corrected				Uncorrected				Bed corrected				
	D	I	K	α95	D	I	K	α95	D	I	K	α95	D	I	K	α95	
10–11	224	48	11	85	254	− 3	—	—	208	6	21	57	209	− 2	17	64	2
13–17	22	−12	19	18	42	−21	12	23	36	−18	25	16	54	−11	15	20	3
18–23	38	−16	30	12	46	6	43	10	37	−25	56	9	54	1	67	8	4
24–28	50	36	28	15	2	30	28	15	58	25	7	31	20	33	8	29	5
29	206	25	10	42	236	22	10	42	214	3	14	35	219	5	14	35	6
30–34	Sample data completely inconsistent																
35–36	4	− 4	93	26	14	−15	93	26	1	− 5	107	24	13	−18	106	25	1

[1] Samples 35 and 36 are of volcanic tuff at base of La Quinta section, and have been demagnetized by alternating fields at 150 oe. See introduction for key to abbreviations.

As can be seen by comparison of Figures 9 and 14 and Table III, the samples collected at locality III can be divided into subgroups which correlate with stratigraphic subdivisions. The statistical data for these individual subgroups are better than the average for all samples (*cf.* Table I).

Samples III-35 and III-36 are from a dacitic tuff unit which constitutes the basal part of the section at the type locality. The remanence is relatively strong and stable to both AF and thermal demagnetization. The within-sample precision is high ($K = 2{,}000$). There is no obvious reason to suspect that the remanence is not primary, acquired at the time of extrusion (if ignimbritic) or deposition of the tuff.

The declination of all redbed samples at this locality is distinctly east of that of the tuff. Furthermore, the average for subgroups III–1-5 is significantly different from that for subgroups III–1-3 and III–1-4. The reason for the inconsistency in the redbed data is not known.

Perhaps an even more striking contrast in the overall data is the fact that, whereas the average northeast declination at locality III has negative inclination and the southwest declination is positive, the opposite is true at locality I. However, this correlation between the average vectors becomes worse if the beds are rotated to a horizontal position. The relative rotation needed to improve the correlation is the opposite of that effected by rotating the bedding to horizontal. Hence, there is a clear implication that, at the time of magnetization, the bedding at one or the other locality (or both) was dipping in the opposite direction. This argument is developed only to justify the conclusion that the magnetism of the La Quinta redbeds was acquired or imposed during the folding, after the earliest rotation and before the last.

Source of magnetism.—By a chemical demagnetization technique, Collinson (1965a, b, 1967) demonstrated that, whereas in some redbeds most of the NRM resides in the "black grains" (specularite), in others it resides in the red pigment. Such data are germane to the question of whether the measured remanence is of detrital or secondary origin. Demonstration that the NRM resides in the "black grains" does not prove that the magnetism is detrital, but proof of its presence in the pigment would be strong evidence of CRM.

Young (1967) investigated the source of the magnetism in some La Quinta specimens. Samples were crushed and screened to −120 +200

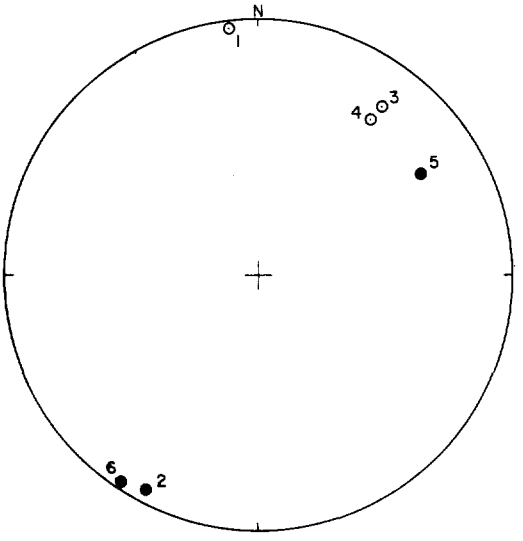


FIG. 14.—Lambert meridional equal-area projection of average demagnetized vector for the individual stratigraphic subgroups at locality III (Table III) before correction for bed attitude. Key to open and closed circles in Figure 9.

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mesh (125–74μ) and the heavy opaque grains separated from the light, red-pigmented residue by bromoform (sp. gr. 2.8). Aliquots of these two fractions and of the whole rocks were reoriented according to the NRM of the individual particles by deposition under gravity in a 30-gauss magnetic field. The sedimenting medium was either wax or silica gel. The remanence was measured (after freezing of the wax or gelation of the gel) after 30-gauss AF demagnetization to remove any magnetism induced during the experiment.

In every test Young found that the specific remanent intensity of the reoriented "whole-rock" powder exceeded that of the uncrushed rock by a factor ranging from 4 to 100. Because a similar increase was found with crushed slate (CRM) and basalt (TRM), it is suspected that this increase is in some way a consequence of strains or dislocations induced during crushing, or is an anhysteritic component acquired by the mechanical deformation (Hargraves and Perkins, 1967). Hence, the magnetic properties of the separated fractions are not necessarily the same as those of the uncrushed rock.

If the weight percentage of the two separated fractions were known, the relative contribution of opaque grains and pigment to the remanence of the whole-rock powder could be determined. In fine-grained samples (123, 95) from which very little heavy mineral was recovered, the remanence appeared to reside mainly in the pigment; in a coarser sample (88) the opaque grains accounted for 98 percent of the remanence.

Thermal demagnetization of the separated fractions showed, in the example of the heavy fraction in sample 123, a complete reversal of sign. This evidence suggests that the remanence of even the individual heavy opaque grains is composite and hence largely of chemical origin, acquired during a significant period of time.

Young found no important difference in the thermal-stability spectra of opaque grains and pigment, which further suggests that even the opaque grains may be largely microcrystalline aggregates, and possibly diagenetic replacements of pre-existing particles. Geologic circumstances favorable for the acquisition of CRM may have been provided by the aforementioned vertical uplift.

Conclusions

1. Of the samples studied, two volcanic tuff specimens from the base of the La Quinta section at the type locality may carry the only unaltered primary Triassic(?) remanence. After correction for the present attitude of the bedding, the average for these samples ($D = 13$, $I = -18$, $K = 106$) gives a south pole at 68.5° S, 108.5° W.

2. The remanence of the redbeds, even after thermal demagnetization, appears to be secondary and to have been acquired during a significant time interval during and/or subsequent to the main period of folding. Hence, no conclusions can be drawn from the paleomagnetic measurements concerning the structural history or the age of the rocks.

3. Samples from localities I and III include both normal and reverse polarizations interspersed irregularly through the section. No time-stratigraphic significance is attached to these alternations; the secondary magnetism is thought to have been acquired during a time interval "straddling" at least one field reversal, but could have been fixed simultaneously at different places in the stratigraphic column, and at different times in different localities, as a result of vagaries of diagenesis.

4. The average of the secondary magnetization of the redbeds has an inclination comparable with that to be expected from the present latitude of Venezuela, but the declination is distinctly east of that expected.

If this magnetism postdates the folding (*i.e.*, is post-Eocene), it implies either clockwise rotation of South America in comparatively recent time, after it arrived in its present latitude (assuming continental drift), or a tectonic rotation associated with the Andean deformation(s). There is no evidence in published paleomagnetic data on South America (Creer, 1964) for the rotation of South America as a whole in Holocene time. A possible mechanism for the tectonic rotation of the Venezuelan Andes was suggested in the discussion of the tectonic setting. However, more unambiguous data are needed before the paleomagnetic method can provide strong support for such an hypothesis.

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